

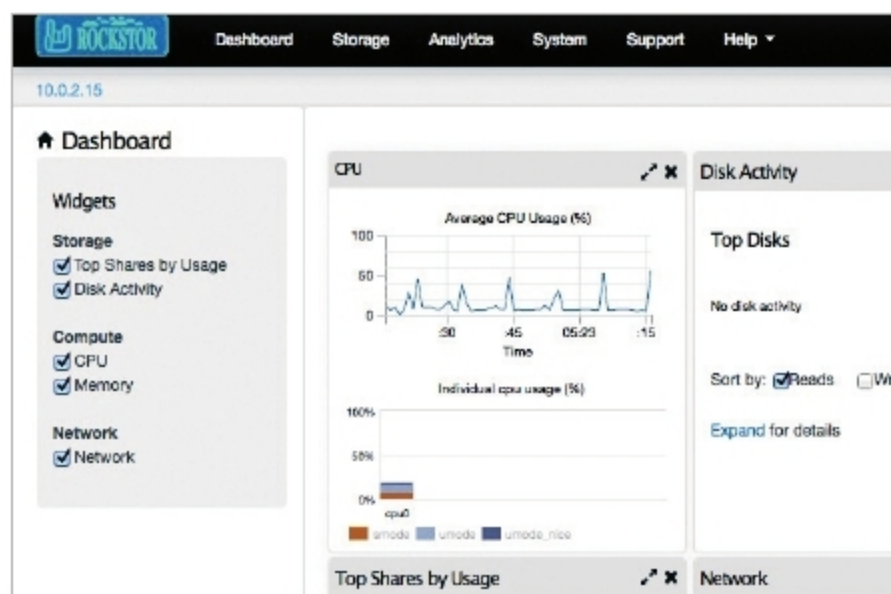


Figure 7: Rockstor

NAS features like file sharing via NFS, Samba, SFTP and AFP, advanced features such as online volume management, CoW (copy-on-write) snapshots, asynchronous replication, compression and bit rot protection are supported based on Btrfs. Rockstor also provides apps like ownCloud, Syncthing, openVPN and Plex to name a few. Apps or 'Rock-ons' are powered by a Docker based application hosting framework. These Rock-ons, combined with advanced NAS features, turn Rockstor into a private cloud storage solution accessible from anywhere, giving users complete control of cost, ownership, privacy and data security.

The Rockstor UI is written in JavaScript, making it simple to manage everything from the Web browser. The back-end is written in Python and exposes RESTful APIs to easily extend functionality.

Features

- It efficiently performs storage management including pool and share creation, snapshots, nfs exports, etc.
- It acts on disk alerts, and repairs and removes disks from the system if necessary, without unnecessary disruption.
- It customises the dashboard per admin user by choosing the relevant widgets and their layout.
- It creates snapshots for shares instantly or schedules snapshots to be created periodically.
- Rockstor NUT reconfiguration aims to make UPS setup easy and straightforward.
- It installs on commodity 64-bit hardware or hypervisors like Xen, VMware and KVM.
- Advanced Btrfs features include dynamic pool management, software RAID, and more.

Latest version: 3.9.1

Official website: <http://rockstor.com>

CryptoNAS

CryptoNAS is a Linux based NAS server designed to encrypt the drives used with it. It can turn almost any old PC into a secure NAS server. CryptoNAS supports almost all operating systems like Linux, MacOS X and other OSs that support Samba shares. It creates encrypted disk partitions using the LUKS volume; the default encryption algorithm is aes-cbc-



Figure 8: CryptoNAS user interface

essiv:sha256. The algorithms are also accessible directly in Linux or directly in Windows via FreeOTFE.

CryptoNAS provides two packages: CryptoNAS-Server and CryptoNAS-CD. The CryptoNAS-Server package adds hard disk encryption to a file server (running Samba, NFS, DAV, etc) without annoying its users. The CryptoNAS-CD is targeted at users who want to set up an encrypting file server without bothering about complicated administration issues.

Features

- It is a multilingual Debian-based Linux live CD with a Web based front-end that can be installed into a hard disk or USB stick.
- It provides tons of choices with regard to encryption algorithms; the default is AES, which means that any Linux operating system can also access them without using CryptoNAS software.
- It has a very user friendly, Web based front-end for hard disk encryption to a file server.

Latest version: 0.3.5

Official website: <http://cryptonas.senselab.org/>

Note: All images are for representation only. 

References

- [1] <https://www.freenas.org/>
- [2] <https://www.xigmanas.com/>
- [3] <https://www.openmediavault.org/>
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- [5] <http://rockstor.com>
- [6] <http://cryptonas.senselab.org/>

 By: Anand Nayyar

The author works at Duy Tan University in Vietnam. He loves to work and research on open source technologies, sensor communications, network security and the Internet of Things. He can be reached at anandnayyar@duytan.edu.vn. YouTube channel: *Gyaan with Anand Nayyar* at www.youtube.com/anandnayyar.